AGENT DESIGN STUDIO

Inpatient claim evidence and adjudication assistance

PASS | DESIGN ONLY

A reviewed system-design specification,not an implemented system or executed business test. Scores below are supplied design judgements.

Review: self-review / round 3 | 87 canonical cards across ten stages.

This report prioritizes design calls,contracts and review evidence. Full fixtures and exact canonical wording remain in design.json; eval-set.json contains the complete machine-readable evaluation set.

Design dimension	Score	Design dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	3/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	3/4	D11	3/4
D6	4/4	D12	3/4

1 Context

Original brief

Design a health insurance claim adjudication assistant for inpatient hospital claims. It should read submitted policy and hospital documents, assess coverage and eligible charges, recommend or process eligible claims, and route exceptions and rejections for review. Policies can change over time and bills may include room-rent limits. Protect personal medical data, explain decisions with evidence and propose any missing rules as assumptions, not verified insurance or medical advice.

CTX1 - Business decision

Business decision: Give adjudicators a traceable inpatient coverage recommendation and exact eligible line amounts. Start on authenticated claim submission; finish with durable review packet, owned hold, cancellation or intake rejection. A recommendation is never insurer approval.

CTX2 - Actors and systems

Actors and systems: Member/hospital submits documents; authenticated intake operator binds member; policy lead curates rule versions; adjudicator decides; privacy officer owns evidence use. Policy registry and claims ledger are authoritative; retrieval and LLM text are supporting evidence only.

CTX3 - Scope and rejected option

Scope and rejected option: One product and currency; extract hospital bill and discharge details, verify policy binding, calculate limits, explain coverage and route. Reject autonomous end-to-end payment/denial because authority, appeals obligations and insurer APIs are unknown. No clinical diagnosis or medical advice.

CTX4 - Inputs and outputs

Inputs and outputs: Inputs: member and tenant identifiers, policy/member coverage, admission/discharge dates, itemized bill, discharge summary, receipts and evidence manifest. Output: versioned recommendation with line calculations, citations, unresolved facts, reviewer queue and immutable provenance. Missing documents permit provisional intake.

CTX5 - Measurement

Measurement: Handling time = adjudicator active minutes per reviewed case; collect 2 weeks baseline before experiment. Correctness = adjudicated exact line amount matches / labelled lines weekly; evidence precision = supported citations / all citations weekly. Safety counts wrong-subject exposures and unauthorized external actions, target zero. Time reduction 20% is a hypothesis, never offsets safety.

CTX6 - Privacy and dependencies

Privacy and dependencies: No real medical data until approved processing/residency/retention and all tenant controls are verified. Working copies proposed 30 days after close with holds; audit identifiers and hashes follow separately approved retention. H1-H6 and H9 block live use.

Evidence boundary: brief quotes and proposed H assumptions are separate. Validation checks source form; it does not establish policy truth or permissions.

2 Task map

T0 - Produce an accountable inpatient recommendation

Tree position: Overall job / sequence / risk high

T1 intake precedes T2 evidence and policy validation, then T3 calculation, then T4 disposition; early failure goes directly to owned hold or cancellation.

Resolution basis: T1 intake precedes T2 evidence and policy validation, then T3 calculation, then T4 disposition; early failure goes directly to owned hold or cancellation.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Authenticated reimbursement submission

T1 - Register and bind a claim revision

Tree position: Child of T0 / leaf / risk high

Authenticate tenant/member, accept provisional evidence manifest, deduplicate exact submissions, detect semantic duplicate candidates without merging.

Resolution basis: Authenticate tenant/member, accept provisional evidence manifest, deduplicate exact submissions, detect semantic duplicate candidates without merging.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Missing discharge summary creates awaiting-evidence case

T2 - Resolve evidence and governing policy

Tree position: Child of T0 / leaf / risk high

Read tenant-bound evidence; extract cited facts; resolve coverage interval and endorsements from registry; discrepancies or missing rules are holds.

Resolution basis: Read tenant-bound evidence; extract cited facts; resolve coverage interval and endorsements from registry; discrepancies or missing rules are holds.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Admission precedes renewal so select historical version only if configured basis says admission

T3 - Calculate and explain eligible charges

Tree position: Child of T0 / leaf / risk high

Code checks dates, member status, remaining benefits, line totals and limits from approved rule pack. Model drafts only evidence grounded recommendation.

Resolution basis: Code checks dates, member status, remaining benefits, line totals and limits from approved rule pack. Model drafts only evidence grounded recommendation.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / INR 6000 room charge against INR 5000 daily cap becomes INR 5000 eligible, not inferred proportional treatment deduction

T4 - Persist owned human disposition

Tree position: Child of T0 / choose one / risk high

Exactly one outcome for a revision: T4.1 review, T4.2 hold, T4.3 cancel. Cancel wins before persistence; uncertain persistence remains owned until reconciled.

Resolution basis: Exactly one outcome for a revision: T4.1 review, T4.2 hold, T4.3 cancel. Cancel wins before persistence; uncertain persistence remains owned until reconciled.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Adjudicator receives supported eligible recommendation

T4.1 - Deliver recommendation or proposed rejection to adjudicator

Tree position: Child of T4 / leaf / risk high

All supported recommendations including zero eligible amount are human-reviewed. Record durable review packet only.

Resolution basis: All supported recommendations including zero eligible amount are human-reviewed. Record durable review packet only.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Coverage exclusion cites clause and proposes rejection; reviewer decides

T4.2 - Hold unsupported or conflicting cases

Tree position: Child of T4 / leaf / risk high

Missing evidence, ambiguity, stale policy, potential duplicate, tool failure or unsupported scope receives reason and assigned owner; never infer denial.

Resolution basis: Missing evidence, ambiguity, stale policy, potential duplicate, tool failure or unsupported scope receives reason and assigned owner; never infer denial.

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Unclear room category or proportional deduction language goes to policy reviewer

T4.3 - Cancel a pending revision safely

Tree position: Child of T4 / leaf / risk high

Authenticated cancellation stops further model work; atomic persistence cancels only unfinalized local revision. Previously delivered records remain auditable and marked superseded; cancellation does not reverse insurer decision.

Resolution basis: Authenticated cancellation stops further model work; atomic persistence cancels only unfinalized local revision. Previously delivered records remain auditable and marked...

Autonomy: Code or human only; model cannot authorize business actions.

Frequency / examples: Per submitted revision; H7 illustrative 1000 cases/month. / Member withdraws before packet delivery

Attribute	Tasks	Authority / loading
A1 Bound identity	T1, T2, T3, T4.1, T4.2, T4.3	Identity gateway: tenantId/memberId/principal/scopes and authenticated cancellation events; do not accept identity from uploaded text. / on demand
A2 Immutable evidence	T1, T2, T3	Evidence vault: docId/version/hash/page/span/media type/subject binding; upload content is untrusted. / on demand

Attribute	Tasks	Authority / loading
A3 Governing policy and remaining cover	T2, T3, T4.1	Versioned policy registry plus member claims ledger: effective intervals, temporal basis, rule hash, remaining benefit as-of sequence; verify again at delivery. / on demand
A4 Revision and owned operation	T1, T4.1, T4.2, T4.3	Case store: caseId/revision/status/opId/requestDigest/recordId/owner/ackAt/dueAt/outbox; use CAS on current revision. / on demand

3 Components

Contracts below retain field names and action boundaries. Long values are compacted; canonical design.json remains authoritative.

C1 - Intake and identity gate

Group / tasks: Code: validation / T1, T4.3

Reject unauthorized requests before reading medical data. Exact key tenant+requestId has a unique digest; same key/different payload is CONFLICT. Semantic match on member/admission/provider/bill number is candidate only, held for human correlation. Cancellation must own member and expected revision. Intake creates no external claim action.

Inputs

principal:{tenantId,memberId,scopes:string[],active:bool}; required, server authenticated

command:{type:submit|cancel,caseId?:uuid,expectedRevision?:int,requestId:uuid}; submit requires policyId,docIds[] (may be empty); cancel requires caseId and expectedRevision

manifest:DocumentRef[]; each tenantId/memberId/docId/version/hash required

Outputs

intake:{status:ready|hold|cancel|reject,caseId:uuid?,revision:int?,reason:enum,duplicateOf:uuid?}

reason:OK|MISSING_DOCS|IDENTITY|DUPLICATE_CANDIDATE|EXACT_REPLAY|UNSUPPORTED|CONFLICT|CANCELLED; exact replay returns original record

Why code: Deterministic identity, policy arithmetic and durable state must remain inspectable and cannot be delegated to model judgement.

Authorization / approval boundary: Authenticated submitter has case:create or case:cancel scope limited to own tenant/member; revision conflicts do not overwrite.

C2 - Evidence and temporal policy reader

Group / tasks: Code: tools / T2

Logical readEvidence and readPolicy APIs filter tenant/member at repository boundary; 10-second timeout, one retry for unavailable, no retry for subject mismatch. Governing policy is selected using approved date basis from H2, not automatically latest. Missing basis, overlapping intervals, conflicting endorsements or unreadable dates produce hold. Effective date boundaries are half-open. Admission/discharge exact dates must be confirmed against cited spans before calculation. Before version selection, a deterministic supported-format parser reads explicit Admission YYYY-MM-DD and Discharge YYYY-MM-DD labels and verifies document subject. It emits date values and exact spans; missing, non-ISO or contradictory date labels hold with MISSING or UNREADABLE/FACT_CONFLICT through C4, without a model deciding temporal applicability. C2...

Inputs

bound:{tenantId,memberId,caseId,revision}; required

docRefs:DocumentRef[]; policyId:string; admissionDate:date?; sourceDocs contain page text and OCR confidence

registry:{versions:[{id,hash,effectiveFrom,effectiveTo,selectionBasis,memberBinding,rulePack}],endorsements:[]}; required authoritative reader

Outputs

evidence:{status:ready|hold,reason:OK|MISSING|UNREADABLE|SUBJECT|AMBIGUOUS_POLICY|UNAVAILABLE,policyVersion?:string,policyHash?:sha256,documents:Span[],rules?:RulePack,ledgerSequence?:int}

Span:{docId,version,page:int,start:int,end:int,text:string,hash:sha256}; all references tenant/member bound

confirmedDates:{admissionDate:ISO date,dischargeDate:ISO date,citations:SpanRef[]}; optional only when hold; required ready. Normalized parser must have exactly one consistent pair.

Why code: Deterministic identity, policy arithmetic and durable state must remain inspectable and cannot be delegated to model judgement.

Authorization / approval boundary: Read-only scoped service with documented medical-data purpose. Never fetch by unscoped docId.

C3 - Structured fact extraction and explanation

Group / tasks: Agents (LLM) / T2, T3

Single bounded extraction call with advisory coverage rationale. Documents are data, never instructions; no tools, no network, no writes. One schema repair retry with same snapshot is allowed; timeout/unavailable abstains. Code checks every required citation and never uses model confidence as approval. Final narrative combines immutable cited clause text and C4 amounts; model cannot invent rules. Inference input hard maximum is 12000 estimated tokenizer tokens including instructions/schema/evidence. If full evidence cannot fit, hold with INSUFFICIENT_EVIDENCE; never truncate bill lines or select a convenient subset. The 50-page/200-line cap is additional, not an assurance of token fit.

Inputs

evidence:Span[]; confirmed policy/rules immutable hash; required

phase:extract_and_explain; max 50 pages and 200 bill lines (H9 proposed support limit), remove direct identifiers before inference

approvedFields:{admissionDate,dischargeDate,diagnosisText,roomClass,billLines:[lineId,category,amountMinor,serviceDate,quantity],coverageClauses}; schema only

Outputs

proposal:{status:supported|abstain,facts:Fact[],coverage:recommend_eligible|recommend_reject|review,evidenceRefs:SpanRef[],uncertainties:string[],explanation:string}

Fact:{field,value,citation:{docId,version,page,start,end}}; coverage is advisory; monetary results must come from C4

modelError:SCHEMA|TIMEOUT|UNAVAILABLE; any uncited/conflicting material fact becomes abstain

EvidenceRef is discriminated: document span {docId,version,page,start,end} or approved policy clause {policyVersion,policyHash,ruleId}; facts use source type matching responsibility. No cross-type citation substitutes for actual dates or amounts.

Authorization / approval boundary: No business action permission; human resolves coverage interpretation and clinical/necessity disputes.

C4 - Rules and money verifier

Group / tasks: Code: validation / T3

Reconcile bill totals exactly, duplicate line IDs reject to hold; missing categories/negative amounts/currency mismatches hold. Day count follows pack; HC1 inclusive days. Room cap min(room charge, cap*covered days) applies only to room lines in HC1. Never infer proportional reductions on related charges. Sub-limits, deductible, coinsurance and remaining cover order must be explicit approved rule program; unknown rule or ambiguity holds. C4 validates extracted numeric/date facts against normalized source values and flags contradictory originals. HC1 synthetic eligibility requires inpatient=true, bound active member interval covering selected admission and diagnosis coverageFlag=covered explicitly established by approved rule pack; excluded flag proposes rejection with zero eligible amount for human review, never final denial. Unknown...

Inputs

proposal:typed C3 output; evidence and policy snapshot required

rulePack:{version,currency,selectionBasis,dayCount,roomCapMinorPerDay,proportionalRule:explicit_none|explicit_formula|unspecified,deductibleMinor,coinsuranceBps,benefitCaps,rounding,remainingCoverMinor}; all approved required

ledgerSequence:int; dates:ISO dates, amounts:int>=0; no floating point money

Outputs

calculation:{status:ready|hold,reason:OK|FACT_CONFLICT|RULE_MISSING|BILL_MISMATCH|DATE|STALE|INSUFFICIENT_EVIDENCE,lines:[{lineId,billedMinor,eligibleMinor,deductionMinor,ruleId,evidenceRefs}],totalEligibleMinor:int?,currency:string,policyHash:sha256,ledgerSequence:int}

verified:{factCitationsValid:bool,totalsReconciled:bool,rulesComplete:bool}; explanation has arithmetic trace

Why code: Deterministic identity, policy arithmetic and durable state must remain inspectable and cannot be delegated to model judgement.

Authorization / approval boundary: Read-only calculation; no reserve or payment authorization.

C5 - Review queue and timer owner

Group / tasks: Code: human triggers / T4.1, T4.2, T4.3

Create assigned queue record atomically with packet. Proposed H6 uses elapsed UTC 4h ack/48h resolution; overdue escalates at deadlines and stays visible, never auto-closes/denies. Human acknowledgment and resolution require scoped role checks and revision CAS. Request-evidence opens a new revision on new upload. Reviewer signs off outside insurer APIs; no notifications are sent by design. Durable queue assignment and timers are part of C6 transaction/outbox, not a separate best-effort action. Acknowledgment/resolution APIs use command kind human_event with actorId, tenant/member/case, expectedRevision, eventId, eventType ack|reviewed|request_evidence, notesRef and UTC time; fresh role checks are required, rejected events never change status. Accepted events are processed through W8-W11 with opId=eventId and full digest, preserving...

Inputs

caseId/revision/tenantId/memberId; reasonCode and immutable packetRef required

trigger:recommendation|exception|cancel|overdue; ownerRole:adjudicator|policy_reviewer|intake_operator|duty_manager

events:{type:ack|resolve|cancel,actorId,scope,expectedRevision,at:UTC}; resolve outcome:reviewed|request_evidence,decisionNotesRef required

Outputs

disposition:{kind:review_pending|hold_owned|cancelled,ownerRole,ackDueAt:UTC,resolveDueAt:UTC,reasonCode,packetRef?}

timer:{at:UTC,eventId,escalateTo}; queue receipts are durable store outbox entries, no email/Slack sends

humanResolution:{status:reviewed|request_evidence|overdue,actorId,at,notesRef}; records outside insurer decision ledger

Why code: Deterministic identity, policy arithmetic and durable state must remain inspectable and cannot be delegated to model judgement.

Authorization / approval boundary: Queue writes internal only; authenticated reviewers restricted by tenant, caseload and medical-data role.

C6 - Case record and audit writer

Group / tasks: Code: tools / T1, T4.1, T4.2, T4.3

Internal logical persistDisposition API performs fresh principal scope/tenant/member checks and CAS in transaction; checks document hashes, policy registry current governing version and ledger sequence for recommendations; version changes require new revision not overwriting. OpId replay with same digest returns same receipt; changed digest is conflict. Durable proof covers record/audit/outbox before committed. Transaction failure is all-or-nothing. No insurer writes exist. Blocked metadata audit uses a separately scoped service identity allowed only requestId/caseId/reason/owner/UTC/opId, no medical content or new recommendation. It never reuses revoked actor scope. Durable ingress receipt exists before work begins; if blocked-event update fails, platform recovery owns original ingress receipt and never reports completion. The...

Inputs

persist:{tenantId,memberId,caseId,expectedRevision:int,opId:uuid,payload:{kind:review_pending|hold_owned|cancelled,reasonCode,ownerRole,packet?,policyVersion?,policyHash?,ledgerSequence?,evidenceHashes:string[]},digest:sha256}; required

Canonical payload is UTF-8 JSON sorted keys/no whitespace; digest SHA256 covers all fields except digest; request identity binds opId and canonical payload

freshIdentityActive:bool from gateway; currentPolicyHash and ledgerSequence read at commit for review_pending; cancel/hold need no policy snapshot

persist discriminator: disposition|human_event; human_event requires eventId,eventType:ack|reviewed|request_evidence,actorId,notesRef and expectedRevision; action-specific medical fields are absent. All fields are digest-bound; fresh reviewer role required.

Outputs

commit:{status:committed|conflict|denied|stale|unknown,recordId?:uuid,revision?:int,opId,digest}; committed requires durable case row+audit+outbox IDs

error:VERSION|AUTH|POLICY_STALE|TIMEOUT_UNKNOWN; known failure returns conflict/denied/stale; transport timeout is unknown

Why code: Deterministic identity, policy arithmetic and durable state must remain inspectable and cannot be delegated to model judgement.

Authorization / approval boundary: Fresh API enforcement permits only internal disposition persistence by workflow identity bound to case; no external finality or money scope.

C7 - Read-only operation reconciler

Group / tasks: Code: tools / T4.1, T4.2, T4.3

Read operation ledger at 1/5/30 minutes, then place durable recovery hold and hourly read until operator resolution. Never blindly resubmit after timeout. not_found is not proof no delayed commit until store reports transaction definitively aborted; service guarantees must be verified H4. Freeze changed revisions/cancellation writes until result known, record intent, then separately authorize new revision. No successful completion without found receipt.

Inputs

lookup:{tenantId,memberId,caseId,opId,expectedDigest}; required

durable recovery identity limited to operation:read and queue escalation after caller session expires

Outputs

reconcile:{status:found|not_found|pending|unavailable,recordId?:uuid,digest?:sha256,revision?:int}; found requires exact digest and identity

recoveryOwner:platform_on_call; nextPollAt:UTC; callback:{eventId,opId,digest,recordId} deduplicated

Why code: Deterministic identity, policy arithmetic and durable state must remain inspectable and cannot be delegated to model judgement.

Authorization / approval boundary: Separate durable read authority; no write or compensation permission.

4 Workflow map

Node / component	To	Guard or terminal disposition
W1 / C1	W2	intake.status=ready [exclusive; first matching predicate; exhaustive enum/default error routes]
W1 / C1	W6	intake.status=hold [exclusive; first matching predicate; exhaustive enum/default error routes]
W1 / C1	W7	intake.status=cancel [exclusive; first matching predicate; exhaustive enum/default error routes]
W1 / C1	W12	intake.status=reject [exclusive; first matching predicate; exhaustive enum/default error routes]
W2 / C2	W3	evidence.status=ready and !cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W2 / C2	W7	cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W2 / C2	W6	evidence.status=hold and !cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W3 / C3	W4	proposal.status=supported and !cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W3 / C3	W7	cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W3 / C3	W6	proposal.status=abstain and !cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W4 / C4	W5	calc.status=ready and !cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W4 / C4	W7	cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W4 / C4	W6	calc.status=hold and !cancelRequested [exclusive; first matching predicate; exhaustive enum/default error routes]
W5 / C5	W8	review disposition created in memory [exclusive; first matching predicate; exhaustive enum/default error routes]
W6 / C5	W8	hold disposition created in memory [exclusive; first matching predicate; exhaustive enum/default error routes]
W7 / C5	W8	cancel disposition created in memory [exclusive; first matching predicate; exhaustive enum/default error routes]
W8 / C6	W9	commit.status=committed [exclusive; first matching predicate; exhaustive enum/default error routes]
W8 / C6	W10	commit.status=unknown [exclusive; first matching predicate; exhaustive enum/default error routes]
W8 / C6	W13	commit.status in {conflict,denied,stale} [exclusive; first matching predicate; exhaustive enum/default error routes]
W9 / C5	TERMINAL	Durable recordId, auditId and outboxId; local status review_pending|hold_owned|cancelled. Never claim payment, approval or rejection finalized.
W10 / C7	W9	rec.status=found with exact expectedDigest [exclusive; first matching predicate; exhaustive enum/default error routes]
W10 / C7	W11	rec.status in {not_found,pending,unavailable} or found digest mismatch [exclusive; first matching predicate; exhaustive enum/default error routes]

Node / component	To	Guard or terminal disposition
W11 / C7	TERMINAL	recovery_pending with opId, digest, recoveryOwner and nextPollAt; no completion or duplicate write.
W12 / C1	TERMINAL	intake_rejected; no PHI fetched and no claim disposition written.
W13 / C6	TERMINAL	blocked_owned; original record preserved, reason and next-review owner visible, no new successful recommendation.

Failure,retry and human controls

failureDisposition [W1]: Unauthorized requests end at W12 without exposing PHI; service unavailability retains ingress receipt in owned intake queue.

timeout [W1, W5, W6, W7, W9, W11, W12, W13]: 30 seconds

retryPolicy [W1, W4, W5, W6, W7, W9, W11, W12, W13]: No retry; failures produce owned hold.

failureDisposition [W2]: Any unavailable/ambiguous/subject outcome maps to hold with exact reason.

timeout [W2]: 10 seconds per read

retryPolicy [W2]: One retry only for unavailable; exhausted -> W6.

failureDisposition [W3]: Schema failure gets one repair; model error exhausted becomes abstain. Untrusted instructions are ignored and flagged.

timeout [W3]: 45 seconds each attempt

retryPolicy [W3]: At most two total calls for schema repair; no timeout retry.

failureDisposition [W4]: Missing rules and ambiguous proportional deductions never become zero eligible amounts.

timeout [W4, W8]: 5 seconds

failureDisposition [W5]: No claimant decision. Packet includes all line amounts, citations, version and unresolved interpretive warnings.

failureDisposition [W6]: Reason assigns intake operator for missing docs; policy reviewer for policy conflict; duty manager for dependency failure; adjudicator for duplicate candidate.

failureDisposition [W7]: Cancellation creates durable local tombstone; no reversal of prior insurer action.

failureDisposition [W8]: Atomic cancelRequested guard supersedes review packet with cancellation only when same subject authorized. Conflict/auth/staleness freezes revision and requires human revalidation.

retryPolicy [W8]: No write retry after unknown; known conflict/denied/stale never retried in revision.

failureDisposition [W9]: Continue H6 timers for review_pending/hold_owned; a human resolution is recorded as a new event with CAS.

failureDisposition [W10]: Read-only recovery continues independently of caller lifetime; no conflicting resubmission until definitive result.

timeout [W10]: 30 minute initial recovery window

retryPolicy [W10]: Reads at 1/5/30 minutes, no writer calls.

failureDisposition [W11]: Platform on-call retains unknown operation; hourly reads continue under operation:read service identity. Alert at 30 minutes, handoff requires receipt or proven abort; cancellation intent waits.

failureDisposition [W12]: Return opaque denial to requester; audit non-sensitive request ID.

failureDisposition [W13]: C6 blocked-event service identity writes minimal operational metadata only under separately scoped append:blocked authority; revoked requester never gets a write. On event-store failure, durable original ingress receipt remains assigned platform recovery and status is blocked_owned; no successful recommendation. New revision needs fresh actor authority.

5 Framework, tools and data

F1 - Single bounded model inside durable code workflow

Category / permission: fit / none

Chosen design: Use one schema-constrained evidence judgement step C3 with code C1/C2/C4/C6/C7 and human queue C5.

Rejected alternatives: Rejected open-ended multi-agent adjudication: unclear authority and harder provenance.

Tradeoffs / verification: Predictable retries, policy arithmetic and human waits outweigh agent flexibility.

Blast radius: One tenant/member/case revision; inference outage causes holds; no insurer financial mutations.

Authority: Scope H1; fresh service and reviewer roles at APIs. Live adapters and policies require H2/H4/H5 sign-off.

Least privilege: Tenant/member row filters; no wildcard medical read; isolated model identity without tools; operation-read recovery identity.

aws: Unverified candidate: AWS object storage + relational DB + key management in approved region; assess isolation, residency, recovery and deletion.

gcp: Unverified candidate: Google Cloud object storage + relational DB + key management; same isolation, residency, recovery and deletion criteria.

azure: Unverified candidate: Azure object storage + relational DB + key management; same isolation, residency, recovery and deletion criteria.

openSource: Proposed self-managed encrypted object store + PostgreSQL + workers; same criteria, greater patch/backup/key-management burden.

F2 - Database state machine and outbox

Category / permission: orchestration / write

Chosen design: Persist workflow state per revision in transactional case DB, enqueue idempotent workers via outbox; restart by opId.

Rejected alternatives: Rejected in-memory chat chain: cannot survive human waits or reconcile timeout. Temporal-like durable orchestrator remains unverified option.

Tradeoffs / verification: Compare DB workers vs durable orchestrator on persisted waits, replay, CAS, timers, audit and operations. DB workers selected for narrow bounded flow; timer/recovery ownership costs remain. Verify choice via crash/replay spike before use.

Blast radius: One tenant/member/case revision; inference outage causes holds; no insurer financial mutations.

Authority: Scope H1; fresh service and reviewer roles at APIs. Live adapters and policies require H2/H4/H5 sign-off.

Least privilege: Tenant/member row filters; no wildcard medical read; isolated model identity without tools; operation-read recovery identity.

Cloud candidate details match F1.

F3 - Separate readers and internal record writer

Category / permission: permission / write

Chosen design: Medical readers scoped tenant/member/case; C3 receives redacted spans and no tools; C6 can only write local dispositions; no insurer credentials.

Rejected alternatives: Rejected sharing broad insurer API credentials with agent: unnecessary payment/denial blast radius.

Tradeoffs / verification: Fresh authorization and digest checks add latency but prevent cross-member and stale packet writes.

Blast radius: One tenant/member/case revision; inference outage causes holds; no insurer financial mutations.

Authority: Scope H1; fresh service and reviewer roles at APIs. Live adapters and policies require H2/H4/H5 sign-off.

Least privilege: Tenant/member row filters; no wildcard medical read; isolated model identity without tools; operation-read recovery identity.

Cloud candidate details match F1.

F4 - Medical vault and authoritative registry

Category / permission: data / read

Chosen design: Separate encrypted original evidence, approved rules registry, case ledger, redacted inference snapshot and non-content telemetry. Retrieval is evidence index, never policy authority.

Rejected alternatives: Rejected single unversioned vector store for all documents and policy: cannot prove effective coverage or identity.

Tradeoffs / verification: Tenant keys, purpose-bound access, immutable references, hash manifests and deletion/hold jobs increase operating burden. Retention/residency/provider contract remain launch blockers H5.

Blast radius: One tenant/member/case revision; inference outage causes holds; no insurer financial mutations.

Authority: Scope H1; fresh service and reviewer roles at APIs. Live adapters and policies require H2/H4/H5 sign-off.

Least privilege: Tenant/member row filters; no wildcard medical read; isolated model identity without tools; operation-read recovery identity.

Cloud candidate details match F1.

F5 - Audit-first human review

Category / permission: architecture / none

Chosen design: All outputs including proposed rejection remain internal assigned review packets; final insurer adjudication and appeals stay with existing authorized workflow.

Rejected alternatives: Rejected auto-processing eligible claims in first release: benefit-reservation atomicity and authority remain unknown.

Tradeoffs / verification: Captures time savings on evidence preparation while avoiding money autonomy; adoption depends on adjudicator trust and clear visual distinction between recommendation and decision.

Blast radius: One tenant/member/case revision; inference outage causes holds; no insurer financial mutations.

Authority: Scope H1; fresh service and reviewer roles at APIs. Live adapters and policies require H2/H4/H5 sign-off.

Least privilege: Tenant/member row filters; no wildcard medical read; isolated model identity without tools; operation-read recovery identity.

Cloud candidate details match F1.

6 Models and cost

Plan / component	Tier / calls	Input / output tokens	USD per million tokens
M1 / C3	standard / 0.84	12000 / 1800	2 input / 8 output

Zero prices mean unpriced,not free. Numeric budgets and branch frequencies are hypotheses unless evidence states otherwise.

M1 - Standard evidence judgement, bounded budget

Budget rationale and fallback: Standard tier is starting hypothesis for cited long evidence and structured output. Small rejected initially for extraction risk; reasoning rejected initially for cost/latency without demonstrated gain. Compare held-out evidence precision, abstention, numeric extraction and 45s p95 on the same dataset before selecting any model. Larger models never change authority.

Fine-tuning condition: Only after prompt/schema/retrieval improvements fail on recurrent labelled errors; require held-out safety nonregression, improved citation/extraction performance, privacy-approved training basis and net quality/cost benefit.

7 Failure modes

Risk	Specific failure	S / O / D	RPN	Linked evaluations
FM1	Cross-member medical disclosure	10 / 3 / 4	120	EV2, EV20, EV23, EV24, EV25
FM2	Wrong temporal policy	9 / 4 / 5	180	EV3, EV4, EV13, EV23, EV24, EV25
FM3	Room-rent deduction error	9 / 5 / 4	180	EV1, EV5, EV6, EV23, EV24, EV25
FM4	Conflicting evidence accepted	8 / 5 / 5	200	EV7, EV8, EV19, EV23, EV24, EV25
FM5	Document prompt injection	9 / 4 / 5	180	EV9, EV23, EV24, EV25
FM6	Duplicate or concurrent disposition	8 / 4 / 4	128	EV10, EV11, EV12, EV23, EV24, EV25
FM7	Unknown persistence misreported complete	8 / 4 / 6	192	EV14, EV15, EV23, EV24, EV25
FM8	Authority revoked at delivery	9 / 3 / 5	135	EV16, EV17, EV23, EV24, EV25
FM9	Human queue abandonment	8 / 5 / 5	200	EV18, EV23, EV24, EV25
FM10	Missing or unsupported case coerced into decision	8 / 5 / 4	160	EV8, EV21, EV22, EV23, EV24, EV25, EV26

Scores and RPN are author-supplied prioritization estimates,not incident probabilities or verified risk acceptance.

FM1 - Cross-member medical disclosure

Effect / cause: Medical data of another member exposed / Unscoped doc lookup or subject mismatch

Prevent / detect: Server tenant/member binding on every read; deny before fetch. / Repository access audit and synthetic cross-tenant canary.

Owned action / gate: Privacy owner; block launch until EV2 and EV20 pass; stop and quarantine on one exposure.

Metric / business KPI: wrong-subject reads count / Privacy incidents

FM2 - Wrong temporal policy

Effect / cause: Wrong benefits or unsupported rejection / Selecting newest policy or missing/overlapping endorsement effective dates

Prevent / detect: Version selection by approved temporal basis; immutable hash; fresh C6 governing policy comparison. / Temporal fixtures and stale-hash monitor.

Owned action / gate: Policy lead; block launch until EV3,EV4,EV13 pass and H2 is verified.

Metric / business KPI: wrong governing policy count / Correct eligible amount

FM3 - Room-rent deduction error

Effect / cause: Underpay recommendation or overstate cover / Inferred proportional deductions, wrong day counts, cap scope or rounding

Prevent / detect: Integer calculation from explicit rule program; unspecified proportional rule holds. / Exact line comparisons and trace inspection.

Owned action / gate: Policy lead; block launch until EV1,EV5,EV6 pass on approved pack.

Metric / business KPI: line amount exact-match rate / Correct eligible amount

FM4 - Conflicting evidence accepted

Effect / cause: False coverage explanation / OCR errors, conflicting originals or fabricated citations

Prevent / detect: Cited normalized values checked against originals; contradictory dates/totals hold; low OCR is unreadable. / Citation span resolver and discrepancy queue.

Owned action / gate: Claims quality lead; EV7,EV8,EV19 required with 98% evidence precision held-out.

Metric / business KPI: unsupported citation rate / Decision rework

FM5 - Document prompt injection

Effect / cause: Rules bypassed or medical data exfiltrated / Bill text instructs agent to ignore policy or send details

Prevent / detect: No tools/network; immutable rule pack; instructions treated as document content; code rejects unsupported fields. / Injection suite and schema violation count.

Owned action / gate: Security owner; EV9 required with zero forbidden effects.

Metric / business KPI: forbidden action attempts accepted / Privacy incidents

FM6 - Duplicate or concurrent disposition

Effect / cause: Repeated review or overwritten cancellation / Exact replay, semantic duplicate, race on revision

Prevent / detect: Request digest idempotency, semantic holds, unique opId and revision CAS. / Unique constraints, race tests and duplicate record count.

Owned action / gate: Platform owner; EV10,EV11,EV12 required, zero duplicate commits.

Metric / business KPI: duplicate disposition count / Adjudicator handling time

FM7 - Unknown persistence misreported complete

Effect / cause: Lost record or duplicate after retry / Timeout after commit and callback duplication/session expiry

Prevent / detect: Durable receipt required; no blind retry; read-only reconciler and freeze conflicting intent. / Operation ledger reconciliation age and unknown state monitor.

Owned action / gate: Platform on-call; EV14 and EV15 required; >30 minute unknown alerts.

Metric / business KPI: unknown operations older than 30m / Review packet delivery completeness

FM8 - Authority revoked at delivery

Effect / cause: Unauthorized medical record write / Scope revoked after intake or changed payload

Prevent / detect: Fresh gateway and tenant/member/payload digest check inside writer; no LLM write scope. / Denied commit audit and role revocation fixture.

Owned action / gate: Security owner; EV16 and EV17 required with zero unauthorized commits.

Metric / business KPI: unauthorized commits / Privacy incidents

FM9 - Human queue abandonment

Effect / cause: Delayed claim and invisible denial-like hold / Queue unowned, SLA timer failure or overload

Prevent / detect: Mandatory owner; durable UTC timers, visible status and escalation without auto-denial. / Age/ack dashboards and timer lag monitor.

Owned action / gate: Claims operations lead; EV18 required and H6 staffing proof; stop new assisted intake when oldest overdue exceeds 48h.

Metric / business KPI: unowned cases and overdue hours / Claim turnaround time

FM10 - Missing or unsupported case coerced into decision

Effect / cause: Unsupported adverse recommendation / Insufficient evidence, unsupported currency/product or cancellation ignored

Prevent / detect: Provisional intake plus reasoned hold; unsupported scope routed; authenticated cancel precedence. / Missing/unsupported/cancel fixtures and disposition audit.

Owned action / gate: Claims product owner; EV8,EV21,EV22 required with zero auto-rejections.

Metric / business KPI: unsupported decisions count / Decision rework

8 Evaluations

These are test specifications,not executed results. Full self-contained fixtures,ordered mocks and inheritance rules are in eval-set.json and design.json. This report shows concrete outcomes and assertion summaries.

EV1 - Supported room-capped recommendation

Layer / links: golden / tasks T1, T2, T3, T4.1 / risks FM3

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; policyVersion: v1; totalEligibleMinor: 2500000; lines: lineId: L1; eligibleMinor: 500000; deductionMinor: 100000 | lineId: L2; eligibleMinor: 2000000; deductionMinor: 0; recordId: record-1; auditId: audit-1; outboxId: outbox-1

Assertion summary: assertions: totalEligibleMinor == 2500000 | lines.L2.deductionMinor == 0 | status == review_pending | recordId and auditId and outboxId exist | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV2 - Wrong tenant isolated before fetch

Layer / links: golden / tasks T1, T2 / risks FM1

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: intake_rejected; reason: IDENTITY; medicalReads: 0; modelCalls: 0; newRecords: 0

Assertion summary: assertions: medicalReads == 0 | newRecords == 0 | reason == IDENTITY | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV3 - Historical coverage beats latest uploaded version

Layer / links: golden / tasks T2, T3 / risks FM2

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; policyVersion: v1; totalEligibleMinor: 2500000

Assertion summary: assertions: policyVersion == v1 | totalEligibleMinor == 2500000 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any

missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV4 - Overlapping policy interval holds

Layer / links: golden / tasks T2, T4.2 / risks FM2

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: AMBIGUOUS_POLICY; ownerRole: policy_reviewer; totalEligibleMinor: none; modelCalls: 0

Assertion summary: assertions: reason == AMBIGUOUS_POLICY | totalEligibleMinor == null | ownerRole == policy_reviewer | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV5 - Unspecified proportional rule cannot infer deductions

Layer / links: golden / tasks T3, T4.2 / risks FM3

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: RULE_MISSING; totalEligibleMinor: none

Assertion summary: assertions: reason == RULE_MISSING | totalEligibleMinor == null | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV6 - At cap inclusive one-day stay

Layer / links: golden / tasks T3 / risks FM3

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; totalEligibleMinor: 2500000; roomDeductionMinor: 0; coveredDays: 1

Assertion summary: assertions: coveredDays == 1 | roomDeductionMinor == 0 | totalEligibleMinor == 2500000 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV7 - Unsupported citation abstains

Layer / links: golden / tasks T2, T3, T4.2 / risks FM4

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: INSUFFICIENT_EVIDENCE; totalEligibleMinor: none

Assertion summary: assertions: reason == INSUFFICIENT_EVIDENCE | totalEligibleMinor == null | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV8 - Provisional intake with missing discharge summary

Layer / links: golden / tasks T1, T2, T4.2 / risks FM4, FM10

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: MISSING_DOCS; ownerRole: intake_operator; modelCalls: 0

Assertion summary: assertions: ownerRole == intake_operator | reason == MISSING_DOCS | modelCalls == 0 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV9 - Bill instructions cannot alter coverage or exfiltrate

Layer / links: golden / tasks T2, T3 / risks FM5

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; totalEligibleMinor: 2500000; networkToolCalls: 0; acceptedInstructionFields: 0

Assertion summary: assertions: networkToolCalls == 0 | acceptedInstructionFields == 0 | totalEligibleMinor == 2500000 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV10 - Exact duplicate returns original receipt

Layer / links: golden / tasks T1, T4.1 / risks FM6

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; reason: EXACT_REPLAY; recordId: record-original; newRecords: 0; modelCalls: 0

Assertion summary: assertions: newRecords == 0 | recordId == record-original | modelCalls == 0 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV11 - Semantic duplicate needs correlation

Layer / links: golden / tasks T1, T4.2 / risks FM6

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: DUPLICATE_CANDIDATE; duplicateOf: case-other; merged: false

Assertion summary: assertions: reason == DUPLICATE_CANDIDATE | merged == false | status == hold_owned | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV12 - Concurrent cancellation wins revision CAS

Layer / links: golden / tasks T4.1, T4.3 / risks FM6

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: blocked_owned; reason: VERSION; newRecommendationRecords: 0; priorCancellationPreserved: true

Assertion summary: assertions: newRecommendationRecords == 0 | priorCancellationPreserved == true | reason == VERSION | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV13 - Policy changes after calculation fail fresh guard

Layer / links: golden / tasks T3, T4.1 / risks FM2

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: blocked_owned; reason: POLICY_STALE; newRecommendationRecords: 0

Assertion summary: assertions: reason == POLICY_STALE | newRecommendationRecords == 0 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV14 - Authorized timeout then found after session expiry

Layer / links: golden / tasks T4.1 / risks FM7

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; recordId: record-1; persistCalls: 1; durableRecords: 1; processedCallbacks: 1

Assertion summary: assertions: persistCalls == 1 | durableRecords == 1 | processedCallbacks == 1 | recordId == record-1 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV15 - Unknown remains owned and cancellation waits

Layer / links: golden / tasks T4.1, T4.3 / risks FM7

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: recovery_pending; recoveryOwner: platform_on_call; persistCalls: 1; completed: false; cancellationWrites: 0; nextPollAt: 2026-10-01T13:30:00Z

Assertion summary: assertions: completed == false | cancellationWrites == 0 | persistCalls == 1 | recoveryOwner == platform_on_call | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV16 - Revoked actor fails commit guard

Layer / links: golden / tasks T4.1 / risks FM8

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: blocked_owned; reason: AUTH; newRecords: 0

Assertion summary: assertions: reason == AUTH | newRecords == 0 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV17 - Changed payload invalidates identity binding

Layer / links: golden / tasks T4.1 / risks FM8

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: blocked_owned; reason: AUTH; newRecords: 0

Assertion summary: assertions: newRecords == 0 | reason == AUTH | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV18 - Human overdue escalates without auto-rejection

Layer / links: golden / tasks T4.1, T4.2 / risks FM9

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-03T12:00:01Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: review_pending; escalations: duty_manager | head_of_claims; ownerRole: adjudicator; autoClosed: false

Assertion summary: assertions: escalations == [duty_manager, head_of_claims] | autoClosed == false | ownerRole == adjudicator | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV19 - Conflicting originals hold regardless of confidence

Layer / links: golden / tasks T2, T3, T4.2 / risks FM4

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1 | discharge-2; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: FACT_CONFLICT; totalEligibleMinor: none

Assertion summary: assertions: reason == FACT_CONFLICT | totalEligibleMinor == null | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV20 - Wrong member document denied

Layer / links: golden / tasks T2 / risks FM1

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: SUBJECT; modelCalls: 0; unboundContentReturned: 0

Assertion summary: assertions: unboundContentReturned == 0 | modelCalls == 0 | reason == SUBJECT | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV21 - Unsupported currency routes to adjudicator

Layer / links: golden / tasks T1, T3, T4.2 / risks FM10

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: hold_owned; reason: UNSUPPORTED; totalEligibleMinor: none

Assertion summary: assertions: reason == UNSUPPORTED | totalEligibleMinor == null | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV22 - Authorized cancellation positive control

Layer / links: golden / tasks T1, T4.3 / risks FM10

Fixture: command: type: cancel; caseId: case-1; expectedRevision: 1; requestId: req-cancel-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; externalPaymentCalls: 0; externalDenialCalls: 0; status: cancelled; recordId: cancel-1; modelCalls: 0; insurerReversalCalls: 0

Assertion summary: assertions: status == cancelled | modelCalls == 0 | insurerReversalCalls == 0 | externalPaymentCalls == 0 | externalDenialCalls == 0; compare: Exact values/enums and record counts; any missing assertion fails. Synthetic mocks are a specification, not executed evidence.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV26 - Supported exclusion is only a proposed rejection

Layer / links: golden / tasks T1, T2, T3, T4.1 / risks FM10

Fixture: command: type: submit; requestId: req-1; caseId: case-1; expectedRevision: 1; policyId: pol-1; docIds: bill-1 | discharge-1; clock: 2026-10-01T12:00:00Z

Expected summary: scope: internal assistance only; status: review_pending; coverage: recommend_reject; totalEligibleMinor: 0; reason: EXPLICIT_EXCLUSION; ownerRole: adjudicator; externalPaymentCalls: 0; externalDenialCalls: 0

Assertion summary: assertions: status == review_pending | coverage == recommend_reject | totalEligibleMinor == 0 | externalDenialCalls == 0 | ownerRole == adjudicator; compare: Exact equality; proposed rejection does not become final denial.

Metric / target: Exact assertion pass fraction and forbidden effects count / All listed assertions true, zero forbidden effects.

Labels / cadence / alert: Policy lead and claims quality reviewer validate synthetic expected outcomes against approved pack before launch; disagreement adjudicated, never label from model output. / Each code/rule/model change before release. / Any forbidden effect blocks release; any assertion regression returns needs-work.

EV23 - Held-out component and end-to-end validation

Layer / links: offline / tasks T1, T2, T3, T4.1, T4.2, T4.3 / risks FM1, FM2, FM3, FM4, FM5, FM6, FM7, FM8, FM9, FM10

Fixture: Versioned policy packs, access-controlled deidentified retrospective claims after H5 and synthetic HC1 suite EV1-EV22. Strata: renewals/endorsements, room caps, unclear proportional rules, OCR, language, missing docs, unsupported scope,...

Expected summary: Exact deterministic totals, correctly supported citations, owned holds, no external financial calls; measure extraction by field and subgroup; replay crash/restart at each persist boundary.

Assertion summary: Run EV1-EV22 assertions independently; zero safety failures; >=99% exact line amounts and >=98% supported citation precision with 95% confidence intervals, subgroup error analysis; proposed H8 thresholds need owner approval. EV26 is also mandatory. For C3 actual-model evaluation ignore ordered model-response mocks, run real bounded inference against each adversarial/missing/conflicting document and validate exact source support/abstention, repeating across random seeds; never report guard mocks as model...

Metric / target: Per-line exact match; field extraction precision/recall; abstention recall; citation precision; crash recovery pass rate. / H8 proposed thresholds; deterministic golden guard suite 100%.

Labels / cadence / alert: Two independent claims reviewers label; policy lead adjudicates disagreements. Evaluate original scanned evidence, never model-labelled truth. / Before model/prompt/rule/adapter release and weekly during pilot. / Any wrong subject, unauthorized action, duplicate commit or false completion blocks release; material subgroup regression blocks expansion.

EV24 - Online safety and queue signals

Layer / links: online / tasks T1, T2, T3, T4.1, T4.2, T4.3 / risks FM1, FM2, FM3, FM4, FM5, FM6, FM7, FM8, FM9, FM10

Fixture: Tenant aggregated access denials, policy hash mismatches, arithmetic discrepancy samples, model schema/abstain counts, duplicate opIds, unknown age and queue timers.

Expected summary: Duty manager sees owned alerts and can disable model-assisted intake while existing queues and reconciliation remain active.

Assertion summary: Page immediately on any confirmed wrong-subject exposure, unauthorized/duplicate commit or false completed record. Alert unknown>30min, ack>4h, resolution>48h; daily blinded quality sample.

Metric / target: Safety event counts; stale-hash rate; duplicate receipt count; overdue/unknown age; unsupported citation audit rate. / Zero confirmed safety incidents; H6 timers and H8 quality guardrails.

Labels / cadence / alert: Claims quality lead validates sampled errors; privacy/security owners classify incidents. / Events continuous; daily quality review; weekly product/operations readout. / Any confirmed safety breach stops new assisted intake immediately. Unknown>30m to platform; queue>48h to head of claims. Resume only after owner-approved repair and replay.

EV25 - Release gate including all canonical goldens

Layer / links: release / tasks T1, T2, T3, T4.1, T4.2, T4.3 / risks FM1, FM2, FM3, FM4, FM5, FM6, FM7, FM8, FM9, FM10

Fixture: requiredGoldenIds: EV1 | EV2 | EV3 | EV4 | EV5 | EV6 | EV7 | EV8 | EV9 | EV10 | EV11 | EV12 | EV13 | EV14 | EV15 | EV16 | EV17 | EV18 | EV19 | EV20 | EV21 | EV22 | EV26; offline: EV23; onlineReadiness: EV24; launchDependencies: H1 | H2 | H3 | H4 | H5 | H6 | H9

Expected summary: Signed release evidence references suite version, every golden result, representative held-out metrics, crash/replay proof and verified policy/privacy/access contracts.

Assertion summary: All EV1-EV22 must pass (new goldens automatically added); EV23 meets approved H8 gates; EV24 alerts exercised; H1-H6/H9 blockers verified; claims/privacy/security/platform owners sign. A single safety failure blocks irrespective of aggregate scores. EV26 is also mandatory.

Metric / target: Required checks passed / required checks; unresolved launch blocker count. / 100% checks and zero unresolved live-use blockers.

Labels / cadence / alert: Named accountable roles approve artifacts; design self-review is excluded as business sign-off. / Every deployment and policy-pack change. / Any missing required result or unresolved blocker prevents real-data rollout.

9 Measurement and rollout

R1 - Business metrics and causal chain

Category / phase / owner: kpi / Baseline and all phases / Claims analytics lead

Business outcome: Active minutes = summed adjudicator engaged minutes per reviewed case; weekly median. Rework = corrected/reopened recommendations / reviewed recommendations within 30 days. No baseline values measured.

AI mechanisms / guardrails: Exact line match protects eligible amount; cite precision reduces rework by exposing correct source. | Abstention recall is safety guardrail, not business benefit. | Median active minutes measures workflow effort; queue age measures operational delay.

Entry / exit / stop and rollback: Collect 2 weeks baseline active minutes using sampled task timers, separate queue elapsed time. Time saved only if evidence preparation is actually removed; 20% median reduction proposed. Track exact line agreement, appeal/rework within 30 days of review and subgroup hold rates. Zero privacy/authority events is hard guardrail.

R2 - Shadow entry and exit

Category / phase / owner: phase / Shadow, at least 2 weeks after dependency verification / Claims quality lead

Business outcome: Median adjudicator active minutes per reviewed eligible inpatient claim, weekly; correctness/rework and privacy safety guardrails.

AI mechanisms / guardrails: Exact line match protects eligible amount; cite precision reduces rework by exposing correct source. | Abstention recall is safety guardrail, not business benefit. | Median active minutes measures workflow effort; queue age measures operational delay.

Entry / exit / stop and rollback: Entry: H1-H6/H9 verified, EV25 executed pass on synthetic data, privacy authorizes real-case shadow. Exit: representative strata labelled through EV23, 99% amount/98% citation proposed gates, no safety events, observed queue capacity. Shadow outputs cannot reach decision UI.

R3 - Assist pilot, terminal scope

Category / phase / owner: phase / Assist pilot 4 weeks plus 30-day rework follow-up / Claims product owner

Business outcome: Median adjudicator active minutes per reviewed eligible inpatient claim, weekly; correctness/rework and privacy safety guardrails.

AI mechanisms / guardrails: Exact line match protects eligible amount; cite precision reduces rework by exposing correct source. | Abstention recall is safety guardrail, not business benefit. | Median active minutes measures workflow effort; queue age measures operational delay.

Entry / exit / stop and rollback: Entry: shadow exit evidence and staffing H6. Randomized reviewer assistance with all packets reviewed; no payment/denial endpoints. Exit: powered time/quality analysis meets approved hypotheses and subgroup safety, review capacity maintained. One safety incident pauses intake; disable feature flag to baseline process, retain records and unknown reconciliation.

R4 - Scale assistant access only

Category / phase / owner: phase / Scale after human go/no-go / Head of claims and privacy owner

Business outcome: Median adjudicator active minutes per reviewed eligible inpatient claim, weekly; correctness/rework and privacy safety guardrails.

AI mechanisms / guardrails: Exact line match protects eligible amount; cite precision reduces rework by exposing correct source. | Abstention recall is safety guardrail, not business benefit. | Median active minutes measures workflow effort; queue age measures operational delay.

Entry / exit / stop and rollback: Entry: assist experiment evidence and owner residual-risk decision. Expand product/volume only after rule-pack validation and held-out strata; autonomy remains assist-only. Stop on any safety failure or overdue queue>48h; restore baseline reviewer UI, preserve audit and complete recovery queues. No autonomous payment phase planned.

R5 - Clustered reviewer assistance experiment

Category / phase / owner: experiment / Baseline then 4-week assist with follow-up / Claims analytics lead

Business outcome: Median adjudicator active minutes per reviewed eligible inpatient claim, weekly; correctness/rework and privacy safety guardrails.

AI mechanisms / guardrails: Exact line match protects eligible amount; cite precision reduces rework by exposing correct source. | Abstention recall is safety guardrail, not business benefit. | Median active minutes measures workflow effort; queue age measures operational delay.

Entry / exit / stop and rollback: No invented sample count. Entry uses shadow feasibility evidence, not completed pilot outcomes. Analytics computes power before enrolling; prolong or decline pilot if insufficient effective sample.

Experiment: Randomize adjudicator teams to assisted vs existing workflow to reduce contamination; stratify by experience and product; distribute eligible cases by fixed routing independent of assignment. Include missing/complex cases for intent-to-treat, log crossovers. Primary outcome continuous active minutes: baseline variance and clinically meaningful effect approved by owner; power two-arm mean difference then inflate by 1+(mean cluster size-1)*ICC, check cluster count and heavy-tail sensitivity/bootstrap CI. Secondary rework binary uses two-proportion power. Analyze all eligible assigned cases, censor unresolved cases transparently and retain safety counts. Four-week cycle plus 30-day rework follow-up; extend if powered sample not reached; no peeking for early benefit.

R6 - Accountable operations and rollback

Category / phase / owner: owner / All phases / Head of claims (proposed accountable role)

Business outcome: Median adjudicator active minutes per reviewed eligible inpatient claim, weekly; correctness/rework and privacy safety guardrails.

AI mechanisms / guardrails: Exact line match protects eligible amount; cite precision reduces rework by exposing correct source. | Abstention recall is safety guardrail, not business benefit. | Median active minutes measures workflow effort; queue age measures operational delay.

Entry / exit / stop and rollback: Claims product owner owns scope and rollout; policy lead owns versioned rules; privacy/security own data and access; platform on-call owns unknown operations; claims operations

lead owns staffing/timers. Only these proposed roles can approve release evidence. No actual role assignments or approvals are claimed.

10 Coding handoff

X1 - Implementation handoff

Acceptance 1: T1/C1 accepts provisional evidence but rejects wrong tenant before PHI access; exact replay returns original receipt and semantic candidate is held.

Acceptance 2: T2/C2 selects approved temporal policy basis with immutable evidence and H2 registry proof; overlapping or missing rules hold.

Acceptance 3: T3/C4 reproduces EV1 INR 2500000 with no proportional deduction on treatment; EV5 unspecified proportional rule holds.

Acceptance 4: C3 has no tools and cannot grant action authority; citations and material facts are verified; one schema repair max.

Acceptance 5: W8/C6 fresh authorization, revision CAS, policy/ledger freshness and full canonical payload binding are enforced atomically; committed requires record/audit/outbox IDs.

Acceptance 6: W10/C7 never retries unknown writes, deduplicates callbacks and retains independent operation-read recovery ownership beyond session expiry.

Acceptance 7: C5 provides owned UTC timers and escalation; all rejection proposals require adjudicator review; no auto-denial, payment or clinical recommendation.

Acceptance 8: EV25 executes EV1-EV22 plus EV23/EV24 readiness; actual external adapter/privacy/policy contracts must verify H1-H6/H9 before live data.

Acceptance 9: EV26 verifies a supported exclusion yields a proposed rejection in review_pending with an adjudicator owner, never a final insurer denial.

Build instructions: Build isolated modules C1-C7 and store Case(tenantId,memberId,caseId,revision,status,owner,timers), EvidenceRef immutable hashes, PolicyVersion/RulePack, Operation(opId,digest,status,recordId), Audit and Outbox. Implement W1-W13 transitions exactly; no LLM agent loop or financial adapter. Use integer minor units, approved rule program and separately versioned prompt/schema. Persist all state with CAS and unique request/op keys; keep no raw PHI telemetry. Provide synthetic mocks from EV1-EV22, component/end-to-end EV23 runner, EV24 alerts and EV25 release manifest. Do not claim these planned tests have executed. Adapter capability, policy interpretation, retention, data-processing authority and owners remain launch dependencies, not coding assumptions. Retain H1 assist-only boundary through R2-R4. Export complete canonical artifacts; render PDF only when requested by parent workflow. After delivering PDF ask exactly once: “Would you also like a visual HTML version in the same ten-stage structure as the app?” Generate HTML only after opt-in. Golden suite also includes EV26; canonical requiredGoldenIds is authoritative, so no added golden is omitted.

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

A Assumptions and launch dependencies

H1 - LAUNCH BLOCKER

Release one recommends eligible amounts and routes all decisions to authorized insurer adjudicators; it cannot issue payment, deny coverage or notify claimants.

Why: Unknown authority and policy interpretation require a bounded useful release.

Proposed owner: Claims product owner

Required validation: Approve assist-only scope and identify adjudicator roles.

H2 - LAUNCH BLOCKER

A tenant policy registry provides legally reviewed immutable policy/endorsement versions, effective intervals, member binding and explicit claim-date selection rules.

Why: Upload text alone cannot establish governing coverage.

Proposed owner: Insurance policy lead

Required validation: Approve temporal rule packs, member coverage lookup and retrospective amendment handling.

H3 - LAUNCH BLOCKER

Synthetic rule pack HC1 uses admission-date selection, inclusive discharge day, INR minor units, room-rent cap only on room lines, no proportional deduction, zero deductible and zero coinsurance.

Why: A concrete executable illustration is needed; none of these are verified insurer rules.

Proposed owner: Insurance policy lead

Required validation: Replace or approve every rule; validate day count, benefit sublimits, exclusions, coordination, remaining sum insured, rounding and temporal basis per product.

H4 - LAUNCH BLOCKER

Internal case store supports tenant scoped transactions, optimistic versions, immutable evidence hashes, durable outbox and operation lookup. All external adapters remain unverified.

Why: Recovery and audit require durable state.

Proposed owner: Platform owner

Required validation: Contract spike proving version CAS, transactional events, idempotency and readback with access controls.

H5 - LAUNCH BLOCKER

Use restricted medical evidence storage, encrypted transport and storage, tenant keys, no provider training, no raw content in logs; proposed 30-day working-copy deletion after case close subject to insurer approved retention/legal holds.

Why: Medical data needs explicit minimization and a verified retention basis.

Proposed owner: Privacy owner

Required validation: Approve lawful basis, residency, model processing contract, access matrix, retention and legal-hold policy before real data.

H6 - LAUNCH BLOCKER

Human queues are staffed by claims operations: acknowledge in 4 elapsed hours, review in 2 elapsed days; overdue goes to duty manager at 4h and head of claims at 48h. Timers run UTC 24x7.

Why: Missing information and exceptions need an owned end state.

Proposed owner: Claims operations lead

Required validation: Confirm staffing, escalation channels, queue capacity and elapsed-time SLA.

H7 - PROPOSED ASSUMPTION

Illustrative volume is 1000 cases/month; 80% reach the model and 5% of reached cases need one schema repair retry. Synthetic standard model prices are USD 2 input and USD 8 output per million tokens.

Why: Provides reproducible capacity/cost hypotheses, not measured provider prices.

Proposed owner: Engineering lead

Required validation: Measure branch mix, token distribution, throughput and approved model quote.

H8 - PROPOSED ASSUMPTION

Proposed release quality gates are zero safety failures in fixed synthetic guard suite, 99% line arithmetic agreement, 98% supported evidence precision and 20% median handling-time reduction.

Why: Explicit hypotheses support go/no-go decisions; do not claim proven benefits.

Proposed owner: Claims product owner

Required validation: Baseline measurement, representative held-out evaluation and sign-off on targets.

H9 - LAUNCH BLOCKER

Scope is one insurer product, one currency per claim and machine-readable or supported scanned inpatient reimbursement claims. Appeals, fraud decisions, outpatient, multi-currency and medical necessity decisions are human-only.

Why: Limits initial policy complexity and avoids pretending to give medical advice.

Proposed owner: Claims product owner

Required validation: Approve product/language/OCR support manifest and exception routing.

B Review evidence and limits

Review method: self-review; round 3; supplied status pass.

Dimension	Score	Supplied evidence
D1	3/4	CTX1-CTX6/T0-T4.3 completely bound supported intake through review/hold/cancel; all final insurer decisions remain human.
D2	3/4	Every high-risk leaf has reachable W coverage and substantive golden. FM/EV reciprocal links derived once; EV26 is included in FM10 and EV25 manifest.
D3	3/4	W1-W13 enumerate ready/hold/cancel/reject and commit/error/unknown paths. Cancellation precedence, CAS and digest readback prevent conflicting resubmission; W11 retains recovery owner.
D4	3/4	C1-C7 declare typed inputs/outputs, read and internal write logical contracts, temporal parser, human-event reentry and SLA roles. Real adapters remain verification dependencies.
D5	3/4	C4 owns integer arithmetic/dates/limits; C6 checks identity/resource/version/full payload at API. C3 has no tools; F3/F4 isolate medical evidence and redacted inference.
D6	4/4	H1-H9 identify every unprovided policy/volume/price/threshold/owner as hypothesis. Sources preserve brief and exact H statements; no clinical/legal claims.
D7	3/4	F1-F5 choose code state machine and bounded single judgement over multi-agent/open-ended adjudication; data candidates labelled unverified against common criteria.
D8	3/4	M1 cost formula .84*$0.0384=$0.032256/case and 2-call worst budget reproducible. Input overflow holds; tier selection requires held-out evidence and fine-tune gates.
D9	3/4	FM1-FM10 contain concrete inpatient-specific effects, causes, S/O/D rationale, owner release gates, control and reciprocal golden coverage. No risk accepted.
D10	3/4	23 self-contained golden fixtures (EV1-EV22,EV26) plus EV23 offline/EV24 online/EV25 release. Exact payload hashes locally checked; EV17 intentionally fails. Actual citation mutation, room cap controls, stale policy, cancellation and callback fixtures inspected. Planned harness remains unexecuted.
D11	3/4	R1-R6 specify unknown-baseline collection, proposed quality/time targets, causal measures, cluster experiment and noncircular shadow/assist/scale gates. Safety event overrides benefits.
D12	3/4	X1 follows canonical C/W/EV IDs; exports generated from same canonical design after frozen validation. Launch blockers remain explicit, no business control execution claimed.

ISSUE5 / major / resolved / evals/EV7

Citation validity was only mocked false although actual model span remained valid; must corrupt a concrete span to isolate guard. Repaired in final canonical version and re-inspected.

ISSUE6 / major / resolved / evals/EV26

Generic fixture augmentation cited last document (policy clause) for admission dates. Correct dates to discharge span and include actual line trace in persist packet. Repaired in final canonical version and re-inspected.

ISSUE7 / major / resolved / evals/EV18

Timer fixture runs at 48h without a clear preexisting committed review record; make event action and prior receipt explicit. Repaired in final canonical version and re-inspected.

Self-review by host model; no independent evaluator or stakeholder sign-off.

Frozen v1 validator verifies structure/references only. No business test harness, insurer integration, medical-data processing or production controls executed.

Synthetic fixtures and illustrative budgets do not prove accuracy, compliance, policy correctness or savings. H1-H6/H9 remain live-use blockers.

PDF is deferred by explicit parent instruction; this run produces canonical JSON and companion exports only.

Design-only tooling: this helper makes no provider, network or business API calls and executes no business assertions; optional host-assistant research is outside this statement.

Schema, reference, graph and ID-coverage checks do not establish semantic completeness, source truth, actual human authorization or production safety.

Review scores and evidence are supplied by the reviewing agent; the helper checks their shape and declared pass conditions, not their truth.

FMEA action/control text is required where specified; the helper does not verify ownership, release authority, regulatory compliance or risk acceptance.

Canonical design SHA256: 5949edad9880af26cf86f1dbe52264af998567104e3ad1d8813e786758a5de52